Programming Problems-Loops 2

What is the index variable in a for loop?
Controls how many times the loop runs
Where does the increment/decrement of the index variable occur?
Happens **automatically at the end** of each loop
What is another name for a for loop?
Counting loop or **counter-controlled loop**.
When are for loops used over while loops in programming?
Use when you **know how many times** to repeat.
How many parts are in the for loop brackets()?
Explain each part.
Initialization – set start value
Condition – test to continue or stop
Increment/Decrement – update counter

Write a program to display the first 10 natural numbers.
Expected Output :
1 2 3 4 5 6 7 8 9 10

```
print("The first 10 natural numbers are:")

for i in range(1, 11):

   print(i, end=" ")

print("\n")
```

Write a program to find the sum of the first 10 natural numbers.
Expected Output :
The first 10 natural number is :
1 2 3 4 5 6 7 8 9 10
The Sum is : 55

```
print("The first 10 natural numbers are:")
```

```
total = 0

for i in range(1, 11):

   print(i, end=" ")

   total += i

print("\nThe Sum is:", total)

print("\n")
```

Write a program to ask a user for a number and display all numbers from 1 to this number and their sum.
User Input : 7
Expected Output :
The first 7 natural number is :
1 2 3 4 5 6 7
The Sum of Natural Number up to 7 terms : 28

```
n = int(input("Enter a number: "))

print("The first", n, "natural numbers are:")

total = 0

for i in range(1, n + 1):

   print(i, end=" ")

   total += i

print("\nThe Sum of Natural Numbers up to", n, "terms:", total)

print("\n")
```

Write a program to read 10 numbers from the keyboard and find their sum and average.
User Input : **the 10 numbers the user provides ex. 1,1,1,1,1,10,10,20,2,3,5**
Expected Output :
The sum of 10 numbers is : 55
The Average is : 5.500000

```
print("Enter 10 numbers:")

numbers = []

for i in range(10):

    num = float(input(f"Number {i+1}: "))

    numbers.append(num)

total = sum(numbers)

average = total / 10

print("The sum of 10 numbers is:", total)

print("The average is:", average)

print("\n")
```

Write a program to display the cube of the number up to an integer the user provides. Example if a user enters 5.

Expected Output :

Number is : 1 and cube of the 1 is :1
Number is : 2 and cube of the 2 is :8
Number is : 3 and cube of the 3 is :27
Number is : 4 and cube of the 4 is :64
Number is : 5 and cube of the 5 is :125

```
n = int(input("Enter an integer: "))

for i in range(1, n + 1):

    print(f"Number is : {i} and cube of the {i} is : {i ** 3}")

print("\n")
```

Write a program to display the multiplication table up to 10 of a given integer.
User Input: **15**
Expected Output :
15 X 1 = 15
...
...
15 X 10 = 150

```
n = int(input("Enter a number to display its multiplication table: "))

for i in range(1, 11):

    print(f"{n} X {i} = {n * i}")

print("\n")
```

Write a program to ask a user for a number and then display this number of odd numbers from 1 and their sum.
Usr Input : **10**
Expected Output :
The odd numbers are :1 3 5 7 9 11 13 15 17 19
The Sum of odd Natural Number up to 10 terms : 100

```
n = int(input("Enter how many odd numbers to display: "))

print("The odd numbers are:", end=" ")

total = 0

odd = 1

for i in range(n):

    print(odd, end=" ")

    total += odd

    odd += 2

print("\nThe Sum of odd Natural Numbers up to", n, "terms:", total)

print("\n")
```

Repeat the last problem but for even numbers.

```
n = int(input("Enter how many even numbers to display: "))
print("The even numbers are:", end=" ")
total = 0
even = 2
for i in range(n):
    print(even, end=" ")
    total += even
    even += 2
print("\nThe Sum of even Natural Numbers up to", n, "terms:", total)
print("\n")
```

User Input **5**
Expected Output :
The even numbers are :2 4 6 8 10
The Sum of even Natural Number up to 5 terms : 30